

Setbkcolor(number). Sets the background colour of choice from 16 colour options (0 through 15). Instead of specifying the number, the respective colour name can also be used in block letters from the specified set.

SetColor(number). Selects a foreground or drawing colour from the same set of 16 colours.

SetTextstyle(font,direction,character size). Sets the values for text characteristics to be used for displaying the text.

Outtextxy(x,y,"Literal expression"). Displays the string in graphics mode. (x, y) denotes the starting coordinate of the string on the screen.

Setfillstyle(pattern,color). Used to fill the colour. It enables an additional feature of filling the colours with specific pattern of colouring as well.

Floodfill(). Used for filling the required colour in geometrical figures.

Geometrical figures can be drawn using the inbuilt functions of shapes (rectangle, circle, etc).

EFY note. The source code of this article has been included in this month's EFY-CD and is also available on www.efymag.com website.

GPS.C

```
// HEADER FILES USED
#include<stdio.h>
#include<conio.h>
#include<graphics.h>

// FUNCTIONS USED
void introduction();
void segments();
void satellite_symbol();
void projectwork();
void proj1();
void proj2();
void proj3();
void proj4();
void closing();

void main()
{
    int gd=DETECT,gm;
    initgraph(&gd,&gm,"c:\\turbo3\\bgi");

    introduction();
    segments();
    satellite_symbol();
    projectwork();
    closing();
    getch();
    closegraph();
    restorecrtmode();
}

/* TO DISPLAY THE NAME OF THE PROGRAMMER */
void introduction()
{
    setbkcolor(RED);
    setcolor(YELLOW);
    delay(1000);
    settextstyle(7,0,6);
    outtextxy(90,50,"GPS System Demo");

    delay(1000);
    setcolor(WHITE);
    outtextxy(240,150,"Project");

    delay(1000);
    setcolor(YELLOW);
    delay(1000);
    settextstyle(7,0,2);
    outtextxy(100,350,"By : Manuj Arora");
    delay(2000);
    cleardevice();
}

/* TO DISPLAY MAJOR SEGMENTS OF GPS */
void segments()
{
    setbkcolor(BLUE);
    setcolor(YELLOW);
    settextstyle(7,0,6);

    delay(1000);
    outtextxy(100,100,"Major Segments");
    delay(500);
    outtextxy(280,200,"OF");
    delay(1000);
    outtextxy(330,300,"GPS");
    delay(3000);
    cleardevice();
    setcolor(WHITE);

    settextstyle(8,0,4);
    outtextxy(100,40,"GPS Segments");
    delay(1000);
    circle(450,120,75);
    setfillstyle(1,LIGHTGREEN);
    floodfill(450,120,WHITE);

    setcolor(RED);
    settextstyle(8,0,2);
    outtextxy(405,100,"CONTROL");
    setcolor(WHITE);
    delay(1000);

    setcolor(YELLOW);
    moveto(375,120);
    lineto(225,225);
    moveto(225,225);
    lineto(245,205);
    moveto(225,225);
    lineto(250,215);
    delay(1000);

    setcolor(WHITE);
    circle(150,250,75);
    setfillstyle(1,CYAN);
    floodfill(150,250,WHITE);

    setcolor(RED);
    settextstyle(8,0,3);
    outtextxy(115,225,"SPACE");
    delay(1000);

    setcolor(YELLOW);
    moveto(222,272);
    lineto(370,365);
    moveto(370,365);
    lineto(350,345);
    moveto(370,365);
    lineto(347,359);
    delay(1000);

    circle(450,370,75);
    setfillstyle(1,LIGHTRED);
    floodfill(450,370,YELLOW);
    setcolor(RED);
    outtextxy(420,350,"USER");
    delay(5000);
    cleardevice();
}

/* SYMBOL USED FOR SATELLITE IN THE PROGRAM */
void satellite_symbol()
{
    setbkcolor(CYAN);
    setcolor(RED);
    delay(1000);
    settextstyle(7,0,4);
    outtextxy(170,40,"SYMBOL USED");
    delay(500);
    outtextxy(275,80,"FOR");
    delay(1000);
    settextstyle(7,0,4);
    outtextxy(215,120,"SATELLITE");
    delay(1000);

    setcolor(BLUE);
    circle(305,250,20);
    setfillstyle(9,RED);
    floodfill(305,250,BLUE);
    setfillstyle(9,BROWN);
    rectangle(255,200,285,300);
    floodfill(260,250,BLUE);

    rectangle(325,200,355,300);
    floodfill(330,250,BLUE);
    delay(2000);
    cleardevice();
}

/* TO DEMO - How GPS WORKS ? ( 3D TRI-
LATERATION ) */
void projectwork()
{
    setbkcolor(BLUE);
    setcolor(WHITE);
    settextstyle(7,0,6);

    delay(1000);
    outtextxy(80,150,"How GPS Works ?");
    delay(2000);
    setcolor(YELLOW);
    outtextxy(80,250,"3D TRI-
LATERATION");
    delay(2000);
    cleardevice();
    proj1();
    proj2();
    proj3();
    proj4();
}

/* TO DISPLAY REGION WHEN 1 SATELLITE
IS CONSIDERED */
void proj1()
{
    setbkcolor(BLUE);
    setcolor(YELLOW);
    settextstyle(7,0,4);
    outtextxy(130,30,"No. of Satellites
Used : 1");
    delay(1000);
    setcolor(WHITE);
    circle(320,240,150);
    setfillstyle(1,YELLOW);
    floodfill(320,240,WHITE);

    setcolor(GREEN);
    setfillstyle(9,BROWN);
    circle(320,240,10);
    floodfill(320,240,GREEN);
    rectangle(290,220,310,260);
    floodfill(300,240,GREEN);
    rectangle(330,220,350,260);
    floodfill(340,240,GREEN);
    moveto(320,240);
    lineto(350,387);
    moveto(350,387);
    lineto(345,376);
    moveto(350,387);
    lineto(350,375);
    delay(1000);
    setcolor(YELLOW);
    outtextxy(100,400,"Region : Imaginary
Sphere");
    delay(3000);
    cleardevice();
}

/* TO DISPLAY REGION WHEN 2 SATELLITES
ARE CONSIDERED */
void proj2()
{
    setbkcolor(BLUE);
    setcolor(YELLOW);

```